

CS & IT ENGINEERING



Computer Network

MAC Sub-Layer

Lecture No. - 03



By - Abhishek Sir



Recap of Previous Lecture



Topic

Pure ALOHA





Topics to be Covered



Topic

Slotted ALOHA

Topic

Back-off Time

ABOUT ME



Hello, I'm **Abhishek**

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Topic : Pure ALOHA



- > When node has packet to transmit, it transmit immediately
- > Allow collision to happen
[recover via “retransmission”]
- > Use randomization in choosing “when to retransmit”
[Back-off time]



Topic : Pure ALOHA



- > All frames are of equal size ✓
- > Receiver send acknowledgement to Source
[for every correctly (without collision) received frame]
- > After transmitting a frame, Source wait for an ACK upto time-out
- > **Time-out time = Round Trip Propagation Delay**



Topic : Pure ALOHA



G : Mean number of transmissions per frame time

[Average number of frame transmitted in one frame transmission time]

S : Throughput per frame time

[Average number of frame transmitted successfully in one frame transmission time]

P = Probability that a frame does not involve in a collision

$$S = G * P$$



Topic : Pure ALOHA

$\Rightarrow$ vulnerable time = $2t_x$



According to Poisson's distribution

$$P = e^{-2G}$$

then

$$S = G * e^{-2G}$$



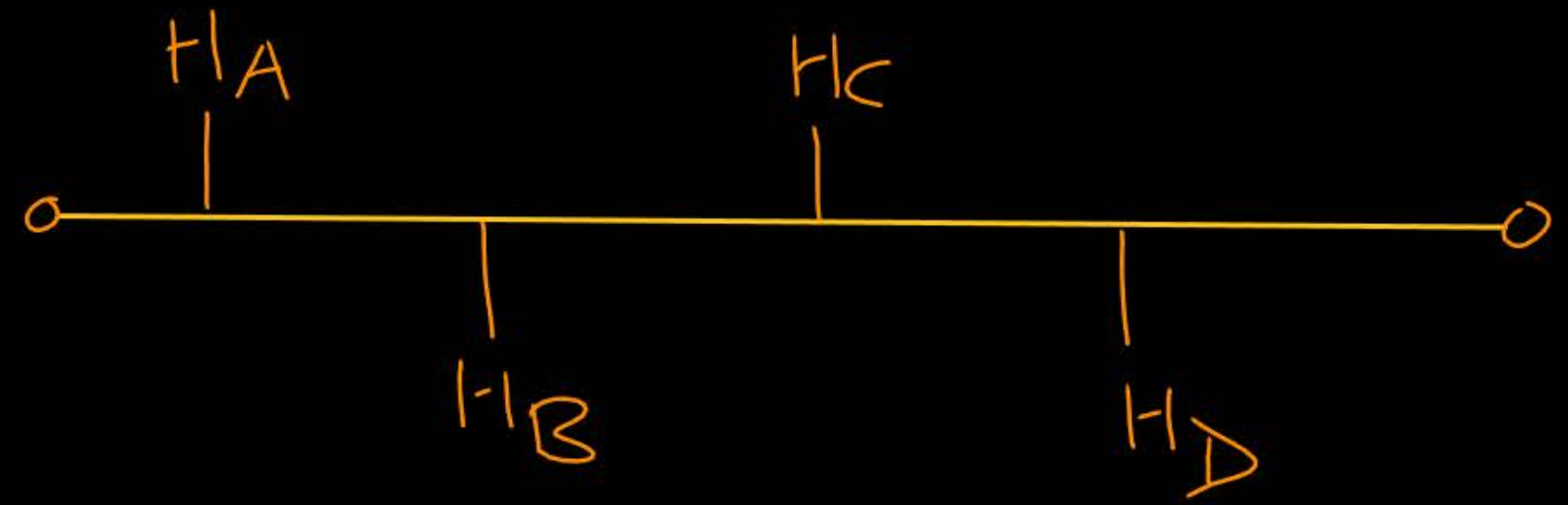
Topic : Pure ALOHA



$$S = G * e^{-2G}$$

Maximum throughput can achieve at $G = 0.5 = \frac{1}{2}$

$$\text{Maximum throughput} = \frac{1}{2}e^{-1} = 0.1839 = 18.39\%$$



#Q. Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1 Mbps ($=10^6$ bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is _____.

$$t_x = \frac{\text{Frame Size}}{\text{Bandwidth}} = \frac{1000 \text{ bits}}{10^6 \text{ bits/sec}} = 10^{-3} \text{ sec}$$

[GATE 2021]

1 sec $\longrightarrow$ 1000 frames

$(t_x = 10^{-3} \text{ sec}) \longrightarrow$ 1 frame

$$G = 1$$

G : Mean no. of transmissions per frame time
 $G=1$



S : Successful no. of transmissions per frame time

$$S = G \times e^{-2G}$$

$$S = e^{-2}$$

$$S = 0.13533$$

$$\approx 0.13717$$

$(t_x = 10^{-3} \text{ sec}) \longrightarrow 0.13533 \text{ frames}$
Successfully transmitted

1 sec $\longrightarrow 135.33 \text{ frames}$

Ans = 135 to 138

IIT-B Ans = 130 to 140



Topic : Slotted ALOHA

- > To improve the efficiency in Pure ALOHA
- > Divide the time into equal size slots
[Slot time = One frame transmission time]
- > Nodes are synchronized
- > Whenever a node has packet to transmit,
it will start transmission only at begin of slot





Topic : Slotted ALOHA



Vulnerable time

-> Time duration in which no transmission should be done to avoid collision

$$\begin{aligned}\text{Vulnerable time} &= \text{Frame Transmission Time} \\ &= t_x\end{aligned}$$

* Slotted ALOHA leads to lesser no. of collisions in comparison to pure ALOHA.



Topic : Slotted ALOHA

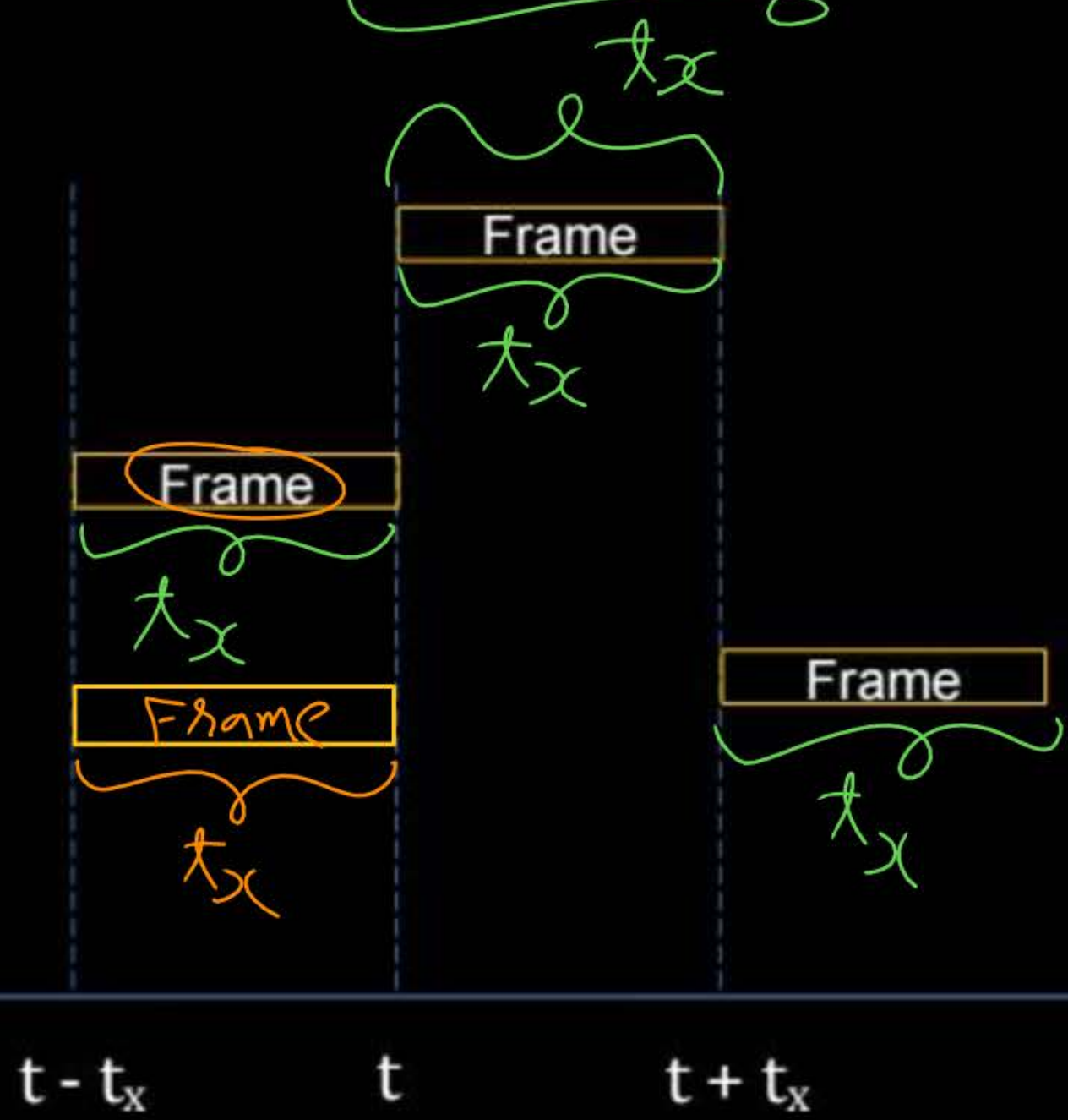
A

B

C

D

vulnerable time



Example 4 :-

Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 2,000 bits. The channel transmission rate is 1 Kbps. Calculate vulnerable time (in seconds) ?

Solution:-

$$\text{Packet Size} = 2000 \text{ bits} = 2 * 10^3 \text{ bits}$$

$$\text{Bandwidth} = 1 \text{ Kbps} = 10^3 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{2 * 10^3 \text{ bits}}{10^3 \text{ bits / sec}} = 2 \text{ Sec}$$

$$\text{Vulnerable time (Slotted ALOHA)} = t_x = 2 \text{ Sec}$$

$$\text{Ans} = 2$$



Topic : Slotted ALOHA

[vulnerable time = t_x]



According to Poisson's distribution ←

$$P = e^{-G}$$



$$S = G * P$$

Then

$$S = G * e^{-G}$$



Example 5 :-

Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 4,000 bits. The channel transmission rate is 2 Mbps. The aggregate number of transmissions across all the nodes with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per frame time. The throughput of the network is _____.

$$t_x = \frac{\text{FrameSize}}{\text{Bandwidth}} = \frac{4000 \text{ bits}}{2 \text{ Mbps}} = \frac{4 \times 10^3 \text{ bits}}{2 \times 10^6 \text{ bits/sec}} = 2 \times 10^{-3} \text{ sec}$$

1 sec $\longrightarrow$ 1000 frames

$(t_x = 2 \times 10^{-3} \text{ sec}) \longrightarrow 2 \text{ frames}$

$$G = 2$$

Slotted ALOHA

$$S = G \times e^{-G}$$

$$= 2 \times e^{-2}$$

$$= 0.27067$$

$$\approx 0.27434$$

Ans =
0.270 to 0.275



Topic : Slotted ALOHA



$$S = G * e^{-G}$$

For what value of G throughput S is maximum?

$$\frac{dS}{dG} = 0$$

$$\begin{aligned} S &= G \times e^{-G} \\ \frac{dS}{dG} &= \frac{d}{dG} [G \times e^{-G}] \\ &= G \times \frac{d}{dG} e^{-G} + e^{-G} \times \frac{d}{dG} G \\ &= G \times e^{-G} \times (-1) + e^{-G} \\ &= e^{-G} \times [1 - G] \end{aligned}$$

$$\frac{dS}{dG} = 0$$

$$e^{-G} \times [1 - G] = 0$$

$$1 - G = 0$$

$$\boxed{G = 1}$$

$$S = G \times e^{-G}$$

$$[\text{put } G = 1]$$

$$S = \frac{1}{e}$$



Topic : Slotted ALOHA



$$S = G * e^{-G}$$

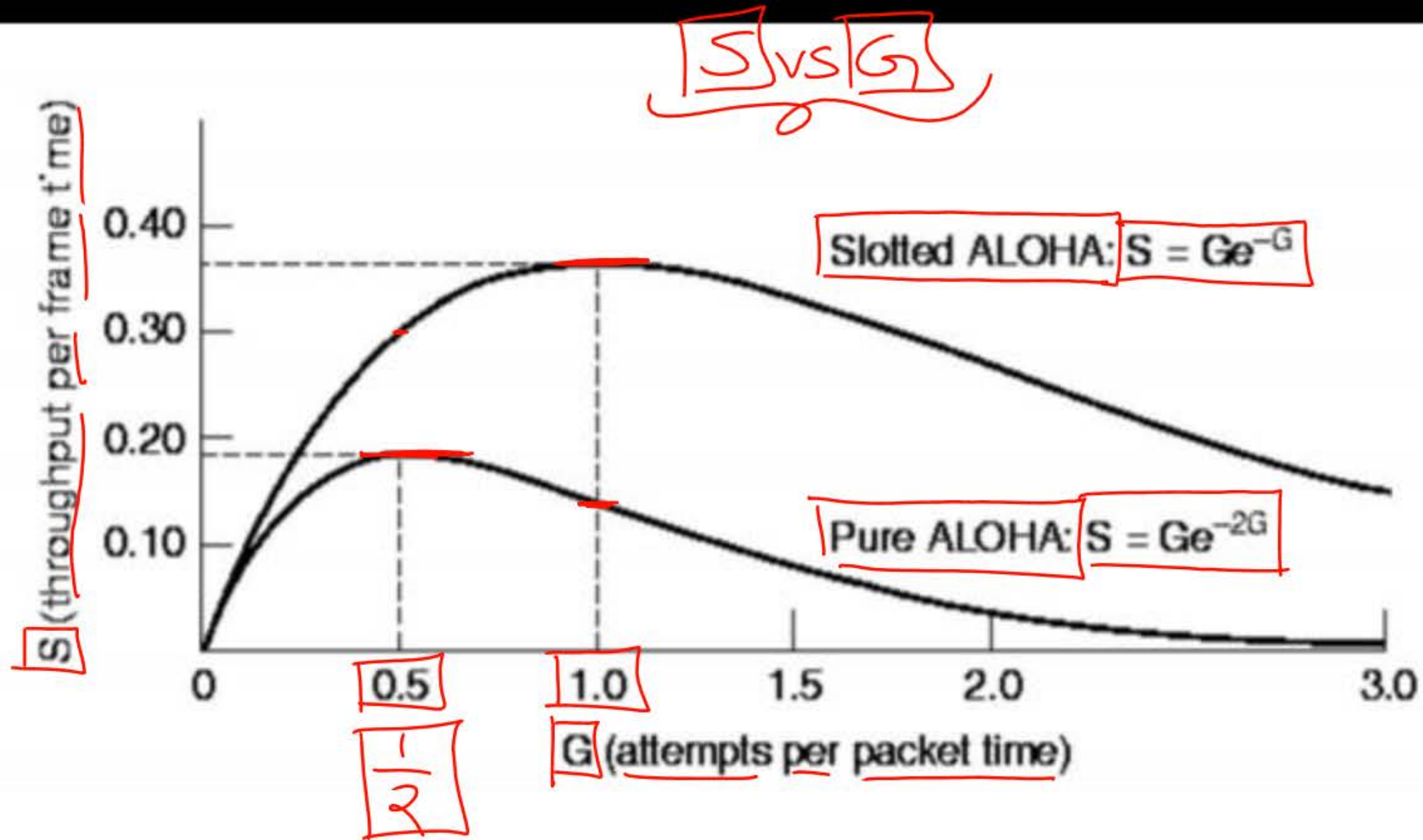
Maximum throughput can achieve at $G = 1$

Maximum throughput = $\frac{1}{e} = 0.3678 = 36.78\%$ ✓

Max^m no. of successful transmission
in one frame time



Topic : Efficiency



#Q. There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that particular one station transmits in a given time slot? *No. of stations = n*

(A) $(1 - p)^{(n-1)}$

H_A

H_B

H_C

H_D

✓ (B) $p * (1 - p)^{(n-1)}$

$$P \times (1 - P) \times (1 - P) \times (1 - P)$$

(C) $n * p * (1 - p)^{(n-1)}$

(D) $1 - (1 - p)^{(n-1)}$

$$\text{Throughput of Host} = P * (1 - P)^{(n-1)}$$

Ans: B

#Q. There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot?

Any one

[GATE 2007]

IIT-K

Throughput of Channel

$$= [\text{Throughput}_A + \text{Throughput}_B + \text{Throughput}_C + \text{Throughput}_D]$$

$$= n * p * (1-p)^{n-1}$$

✓ (A) $n * p * (1-p)^{n-1}$

(B) $(1-p)^{n-1}$

(C) $p * (1-p)^{n-1}$

(D) $1 - (1-p)^{n-1}$

Ans: A



Topic : Slotted ALOHA Efficiency

- > Suppose N nodes with many frames to transmit
- > Each node transmits in slot with probability p

$$0 < p \leq 1$$

Probability that particular node has success in a given slot = $p * (1 - p)^{(N-1)}$
(Throughput of Host)

Probability that any node has success in a given slot = $N * p * (1 - p)^{(N-1)}$
(Throughput of Channel)



Topic : Slotted ALOHA Efficiency

$$\text{Throughput of Channel (A)} = N * p * (1 - p)^{(N-1)}$$

For what value of p throughput A is maximum ?

$$\frac{dA}{dp} = 0$$

$$\frac{d}{dp} [N \times p \times (1-p)^{(N-1)}] = 0$$

$$N \times \left[p \times \frac{d}{dp} (1-p)^{(N-1)} + (1-p)^{(N-1)} \times \frac{d}{dp} p \right] = 0$$

$$\left[p \times (N-1) \times (1-p)^{(N-2)} \times (-1) + (1-p)^{(N-1)} \right] = 0$$

$$(1-p)^{(N-1)} \times \left[p \times (N-1) \times \frac{1}{(1-p)} \times (-1) + 1 \right] = 0$$

$$1 - \frac{p \times (N-1)}{(1-p)} = 0$$

$$1 - \frac{p \times (N-1)}{(1-p)} = 0$$

$$p \times (N-1) = 1-p$$

$$pN - p = 1 - p$$

$$pN = 1$$

$$\boxed{p = \frac{1}{N}}$$



Topic : Slotted ALOHA Efficiency



$$\text{Efficiency} = N * p * (1 - p)^{(N-1)}$$

Maximum efficiency can achieve at $p = 1/N$

$$\text{Maximum efficiency} = \boxed{1/e}$$

$$\begin{aligned} \text{Throughput of Channel} &= N * p * (1 - p)^{(N-1)} \\ &= N * \frac{1}{N} * \left(1 - \frac{1}{N}\right)^{(N-1)} \\ &= \lim_{N \rightarrow \infty} \left(1 - \frac{1}{N}\right)^{(N-1)} \\ &= \boxed{\frac{1}{e}} \end{aligned}$$

#Q. Consider a simplified time slotted MAC protocol, where each host always has data to send and transmits with probability $p = 0.2$ in every slot. There is no backoff and one frame can be transmitted in one slot. If more than one host transmits in the same slot, then the transmissions are unsuccessful due to collision. What is the maximum number of hosts which this protocol can support, if each host has to be provided a minimum through put of 0.16 frames per time slot?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

[GATE 2004]

IIT-D, H.W.

#Q. Consider a LAN with four nodes S1, S2, S3 and S4. Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmit in the same slot. The probabilities of generation of a frame in a time slot by S1, S2, S3 and S4 are 0.1, 0.2, 0.3 and 0.4, respectively. The probability of sending a frame in the first slot without any collision by any of these four stations is _____.

[GATE 2015]

II T - R, (H.W)



2 mins Summary



Topic

Slotted ALOHA ✓

Topic

~~Back-off Time~~



THANK - YOU

